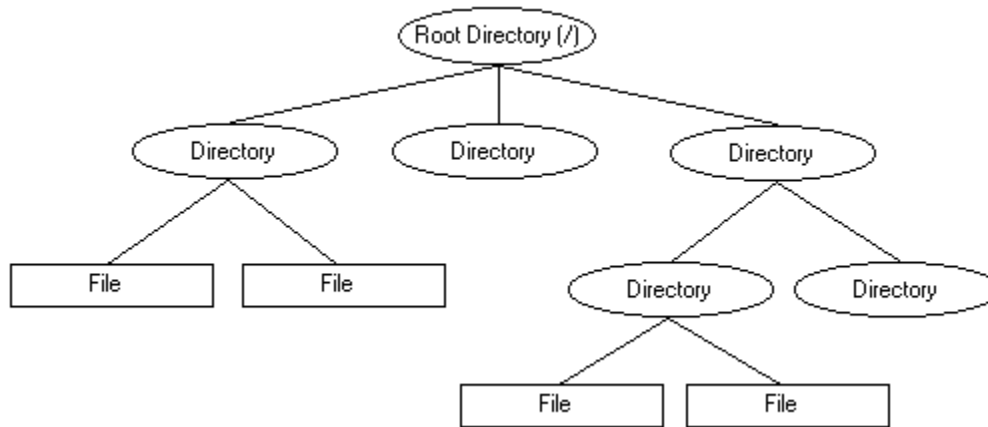
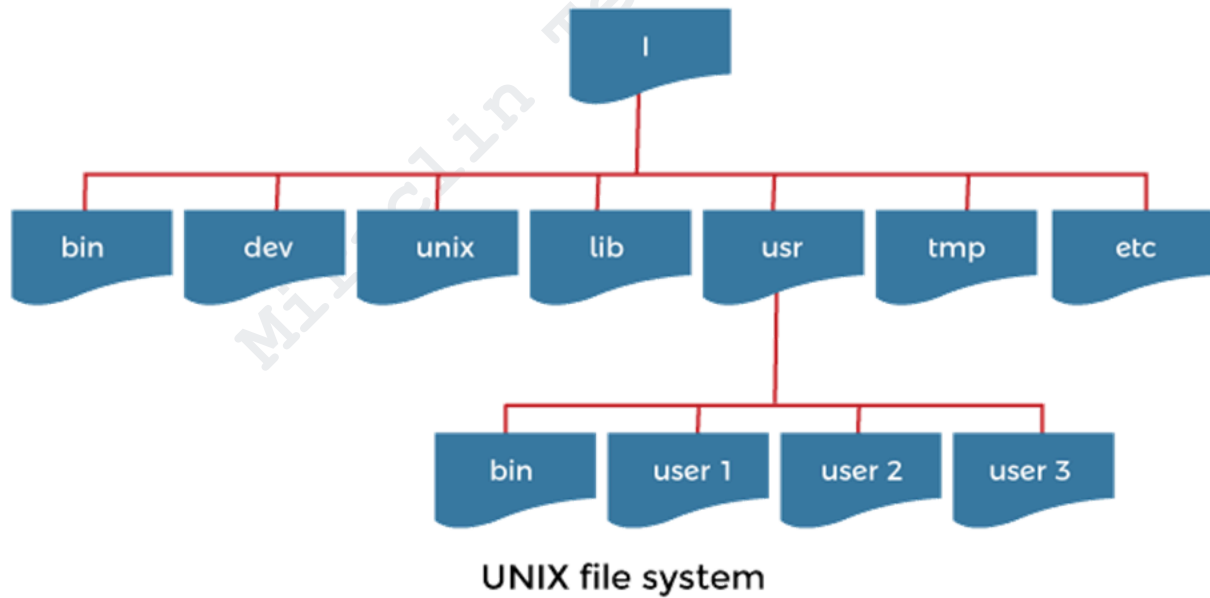


File System Organization



Unix File System Distribution



Types of Files

Types of Files in Linux:

In Linux everything is treated as File.

All files are divided into 3 types

1) Normal or Ordinary files: These files contain data. It can be either text files (like abc.txt) OR binary files (like images, videos etc).

2) Directory Files: These files represent directories. In windows, we can use folder terminology where as in linux we can use directory terminology. Directory can contain files and sub directories.

3) Device Files: In Linux, every device is represented as a file. By using this file we can communicate with that device. Note: short-cut commands to open and close terminal

- `ctrl+alt+t` → To open terminal
- `ctrl+d` → To close Terminal

Creating Files

- Touch Command (Create empty files)-for creating normal or ordinary files.

Eg – `touch sample1`

→ When creating multiple files – `touch sample1 sample2 sample3`

→ When displaying a particular file – `cat filename`

→ When listing (whether the file is present or not)- `ls -l filename`

- `Cat > sample`

Write the text here

And then press CTRL d

- Vi editor

`Vi filename`

Indulging in File Play and directory play

- **Copy a file**

cp file1 file2(This will copy the contents of file1 to file2 .If file2 does not exist it will be created automatically).

- **Copy multiple files in 1 file**

cp file1 file2 file3

- **Removing a file**

rm -i filename (-i option removes files interactively i.e you are asked for confirmation before deleting the files).

- **Removing a directory**

rm -r dir1 (This command recursively removes all the contents of dir1 and also dir1 itself).

rm -rf dir1 (-f option is for forcefully removal).

- **Moving a file(Renaming a file)**

mv test sample

- **Making a directory**

mkdir dir1

- **Removing a directory**

rmdir dir1

- **Change directory**

cd dir1

cd /usr/local/bin

Changes to the directory /usr/local/bin.

From this directory, you can cd to the directory /usr/home/amrood

using the following relative path –

cd ../../home/amrood

Listing files and Directories

The ls command is used to list files or directories in Linux and other Unix-based operating systems. Just like you navigate in your File explorer or Finder with a GUI, the ls command allows you to list all files or directories in the current directory by default, and further interact with them via the command line.

- List files in current working directory → ls
- List files in another directory → ls dirname
- List files in root dir → ls
- list the contents in the users's home directory → ls ~
- List files recursively → Type the ls -R command to list all files and directories with their corresponding subdirectories down to the last file.
- List files with their sizes → Type the ls -s command (the s is lowercase) to list files or directories with their sizes.
- List files in long format → Type the ls -l command to list the contents of the directory in a table format with columns including:
 1. content permissions
 2. number of links to the content
 3. owner of the content
 4. group owner of the content
 5. size of the content in bytes
 6. last modified date / time of the content
 7. file or directory name
- list files including hidden files → ls -a
- list files and output the result to a file → ls > output.txt
- Display all files and directories in reverse of alphabetical order → ls -r
- ls|more → To display content line by line (To come out we have to use q)

- `ls | pg` → To display content page by page. Each page contains 20 lines of content. (To come out we have to use `q`)
- `ls -t` → To display all files based on last modified date and time. Most recent is at top and old are at bottom.
- `ls -rt` → To display all files based on reverse of last modified date and time. Old files are at top and recent files are at bottom.
- `ls -a` → `a` means all. To display all files including hidden files. Here `.` and `..` also will be displayed.
- `ls -F` → To display all files by type.
 1. directory `/`
 2. executable file `*`
 3. link file `@`
 4. Eg: `initctl@` Link File
 5. `pts/` Directory
 6. `ls*` Executable File
- `ls -i` → To display all files including inode number.
- `ls -h` → display in human readable format.

File System Navigation Commands:

1) Every directory implicitly contains 2 directories `.` and `..`

- `.` Represents Current Directory
- `..` Represents Parent Directory

2) `$ cd .` → Changes to Current Directory (Useless)

3) `$ cd ..` → Changes to Parent Directory

4) `$ cd` → If we are not passing any argument, then changes to user home directory.

5) `$ cd ~` → `~` Means User Home Directory. It will Change to User Home Directory. (`$HOME`)

6) \$ cd - → - Means Previous Working Directory. It will change to Previous Working Directory.

2. tac Command:

It is the reverse of cat.

It will display file content in reverse order of lines. i.e first line will become last line and last line will become first line. This is vertical reversal.

```
durga@durga-VirtualBox:~$ cat abc.txt
```

CAT

RAT

MAT

```
durga@durga-VirtualBox:~$ tac abc.txt
```

MAT

RAT

CAT

3. Rev Command:

rev means reverse. Here each line content will be reversed. It is horizontal reversal.

```
VirtualBox:~$ cat abc.txt
```

CAT

RAT

MAT

```
VirtualBox:~$ rev abc.txt
```

TAC

TAR

TAM

Note: cat command will display total file content at a time. It is best suitable for small files. If the file contains huge lines then it is not recommended to use cat command. We should go for head, tail, less and more commands.